Efficient Online Compression for RAG

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Outline

1) **Introduction**
    1) About myself
    2) RAG
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2) **Plug & Play Model Provence**
    1) Architecture
    2) Training
    3) Joint Reranker and Pruner
    4) Results

3) **Soft Compression with PISCO, OSCAR**
    1) Soft Compression
    2) Architecture
    3) Training
    4) Results

About myself

- Phd in Information Retrieval ~ 2010
- …
- NLP, ML Various Projects
- …
- Naver Labs Europe, 2017
- Visit Naver Machine Translation team in South Korea (Papago) in 2018
- Back to IR, Phd supervision, Neural Search → SPLADE

- **Now:**
  - Principal Scientist & Team Lead of Retrieval and Memory team at NLE
    Retrieval, RAG, Memory, LLMs, Agents

Compression for RAG: The challenge

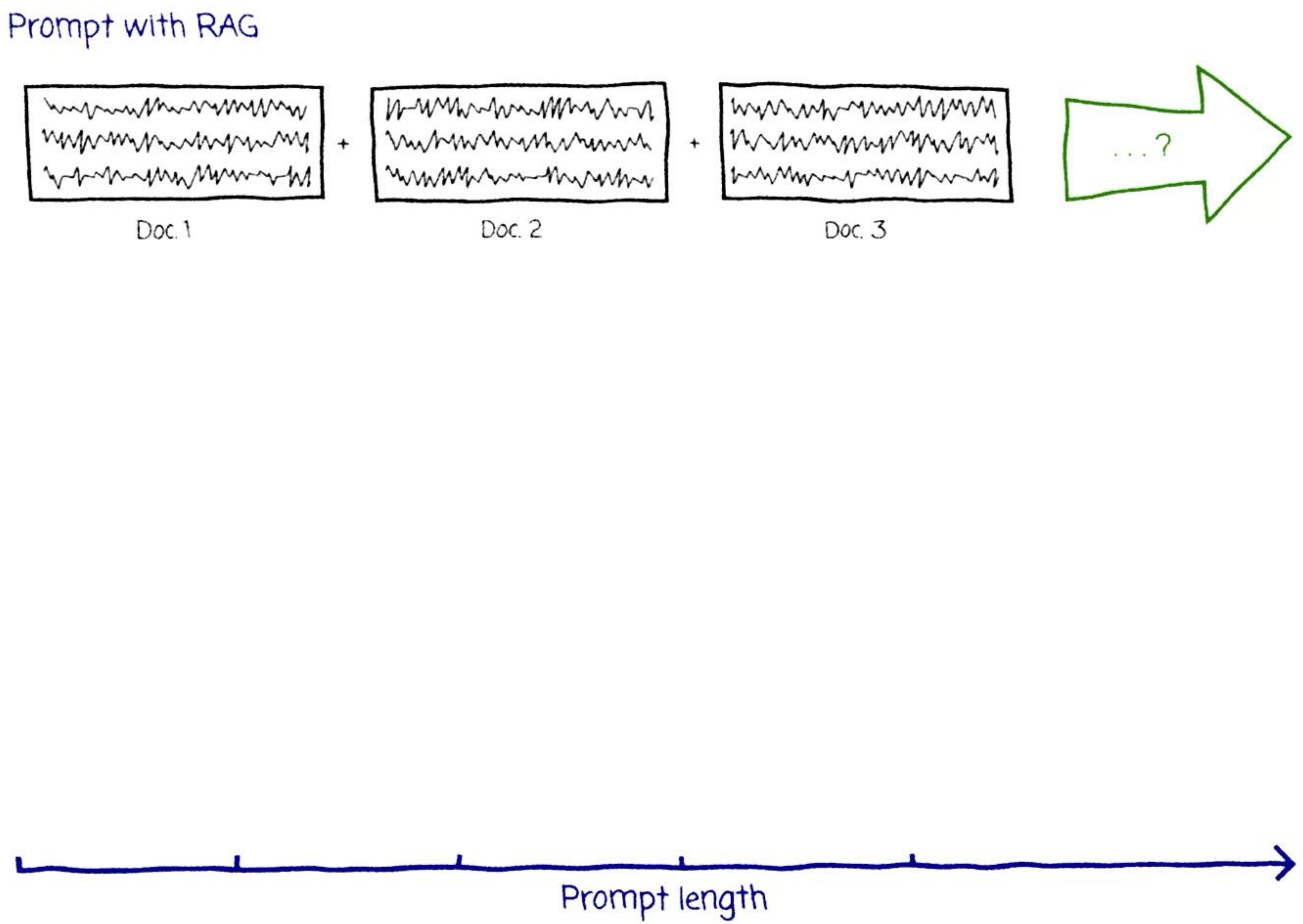
Compression is easy,
maintaining the same level of performance
while improving speed is hard!

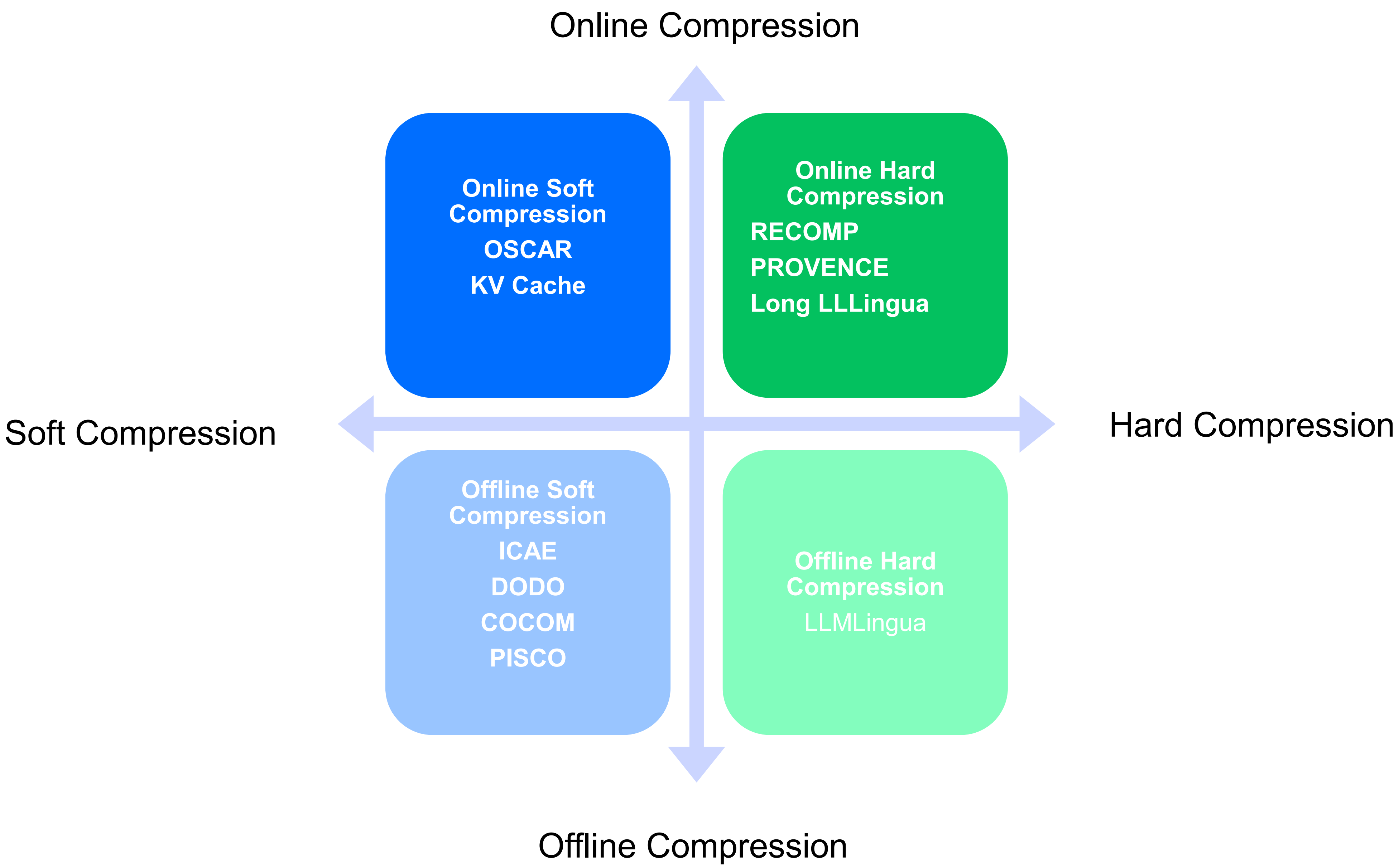
We have designed a **FAMILY** of compression models

- **Plug & Play:** works with any LLM → **Provence**
- **Fine-tuned LLMs** give better performance → **OSCAR**

Disclaimer: Slides, Graphics done by Nadia, Maxime, NLE Communications!

Hard and Soft Compression




Online Compression
Online Soft Compression
OSCAR
KV Cache
Online Hard Compression
RECOMP
PROVENCE
Long LLLingua
Soft Compression
Hard Compression
Offline Soft Compression
ICAE
DODO
COCOM
PISCO
Offline Hard Compression
LLMLingua
Offline Compression

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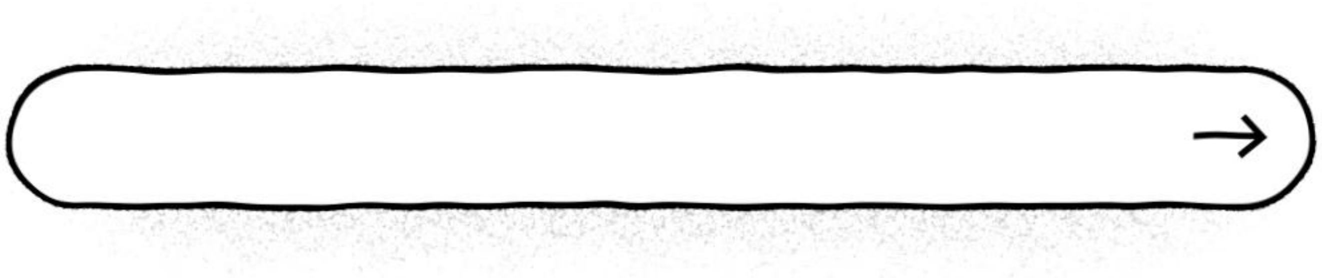
PROVENCE

Efficient context pruning for RAG

Acccepted to ICLR 2025
Paper: https://arxiv.org/abs/2501.16214
Model: https://huggingface.co/naver/provence-reranker-debertav3-v1
https://huggingface.co/blog/nadiinchi/provence
Acronym: *Pruning and Reranking Of retrieVEd relevaNt ContExts*

PROVENCE



Online Compression: Hard Compression

PROVENCE

Question: Can you eat pumpkin every day?

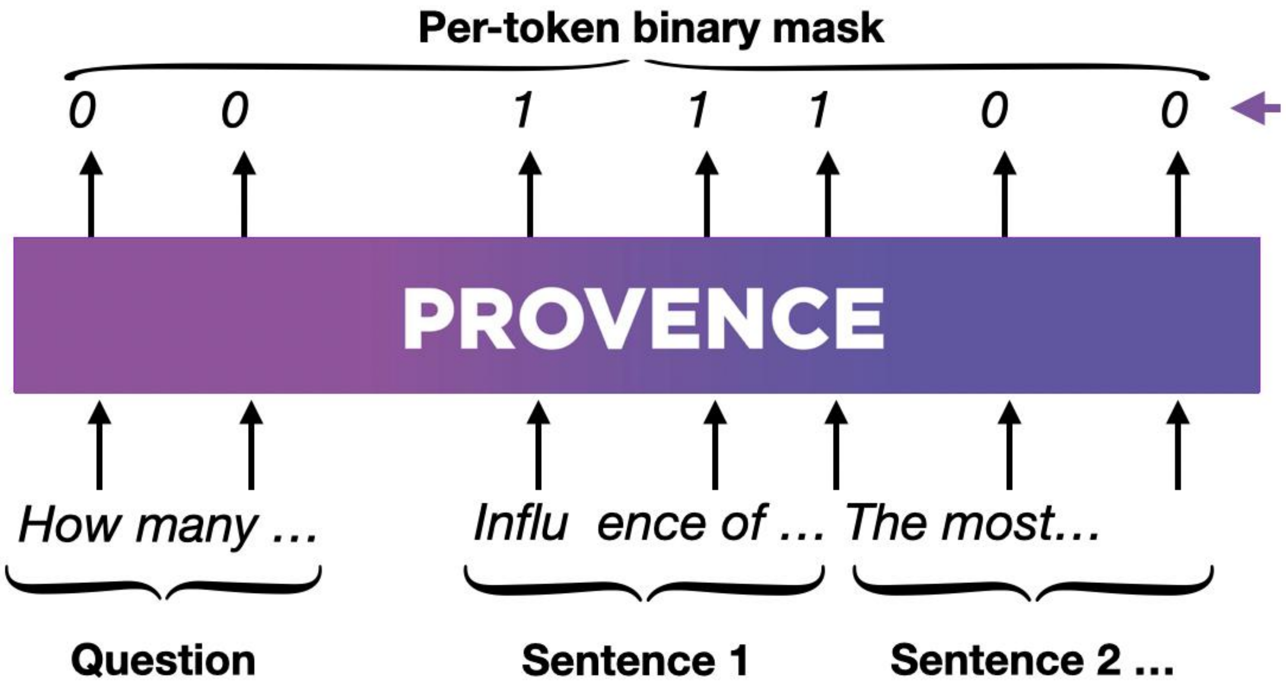
Retrieved context: ~~Pumpkin is at its peak in the fall.~~ Eating pumpkin every day can help reduce inflammation, strengthen your immune system and promote a health. It may also help lower blood pressure. However, if you eat too much, you may experience diarrhea from a high dose of fiber. Read ~~on to learn all about pumpkin's nutrition and health benefits.~~

Relevance score: 0.96

Provence removes sentences that are irrelevant to the user question

…and also outputs a relevance score

Architecture: Pruning as a sequence labelling problem



1s for the tokens of relevant sentences and 0s for all other tokens

Related Works: Ranking sentences, Generation with LLM ..

Training Procedure

Train question

How many French words are there in English?

Retrieved context

Influence of French on English. The most notable influence of French on English has been its extensive contribution to the English lexicon. It has been estimated that about a third of the words in English are French in origin. Linguist Henriette Walter claims that this total may be as high as two thirds. Linguist Anthony Lacoudre has estimated that over 40,000 English words come directly from French.

Number sentences

[1] Influence of French on English. [2] The most notable influence of French on English has been its extensive contribution to the English lexicon. [3] It has been estimated that about a third of the words in English are French in origin. [4] Linguist Henriette Walter claims that this total may be as high as two thirds. [5] Linguist Anthony Lacoudre has estimated that over 40,000 English words come directly from French.

Prompt a strong LLM

Question: *How many French words are there in English?*
Context: [1] *Influence of French on English*. [2] *The most notable influence of French has …*

Answer the Question, using only information provided in the Context. If no useful information is provided, you must output "No answer". **If some parts of the Context are used to answer, you must cite all the corresponding sentences using symbols [ ].**

LLM answer

There are around 40000 French words in English [5], or a third of all words [3].

Silver labeling which sentences are relevant to the given question

Influence of French on English. The most notable influence of French on English has been its extensive contribution to the English lexicon. It has been estimated that about a third of the words in English are French in origin. Linguist Henriette Walter claims that this total may be as high as two thirds. Linguist Anthony Lacoudre has estimated that over 40,000 English words come directly from French.

Optimal Compression depends on the query

Context:

Pumpkin is at its peak in the fall. Eating pumpkin every day can help reduce inflammation, strengthen your immune system and promote eye health. It may also help lower blood pressure. However, if you eat too much, you may experience diarrhea from a high dose of fiber. Read on to learn all about pumpkin's nutrition and health benefits.

Possible questions:	*# relevant sentences:*
Can you eat pumpkin every day?	→ 3 relevant sentences
When is the pumpkin season?	→ 1 relevant sentence
Which vitamins does pumpkin contain?	→ 0 relevant sentences

Provence automatically detects how many sentences are relevant

Key Features

Provence encodes all sentences in the passage and the user question together:
- this enables capturing of coreferences between sentences and provides more accurate context pruning.

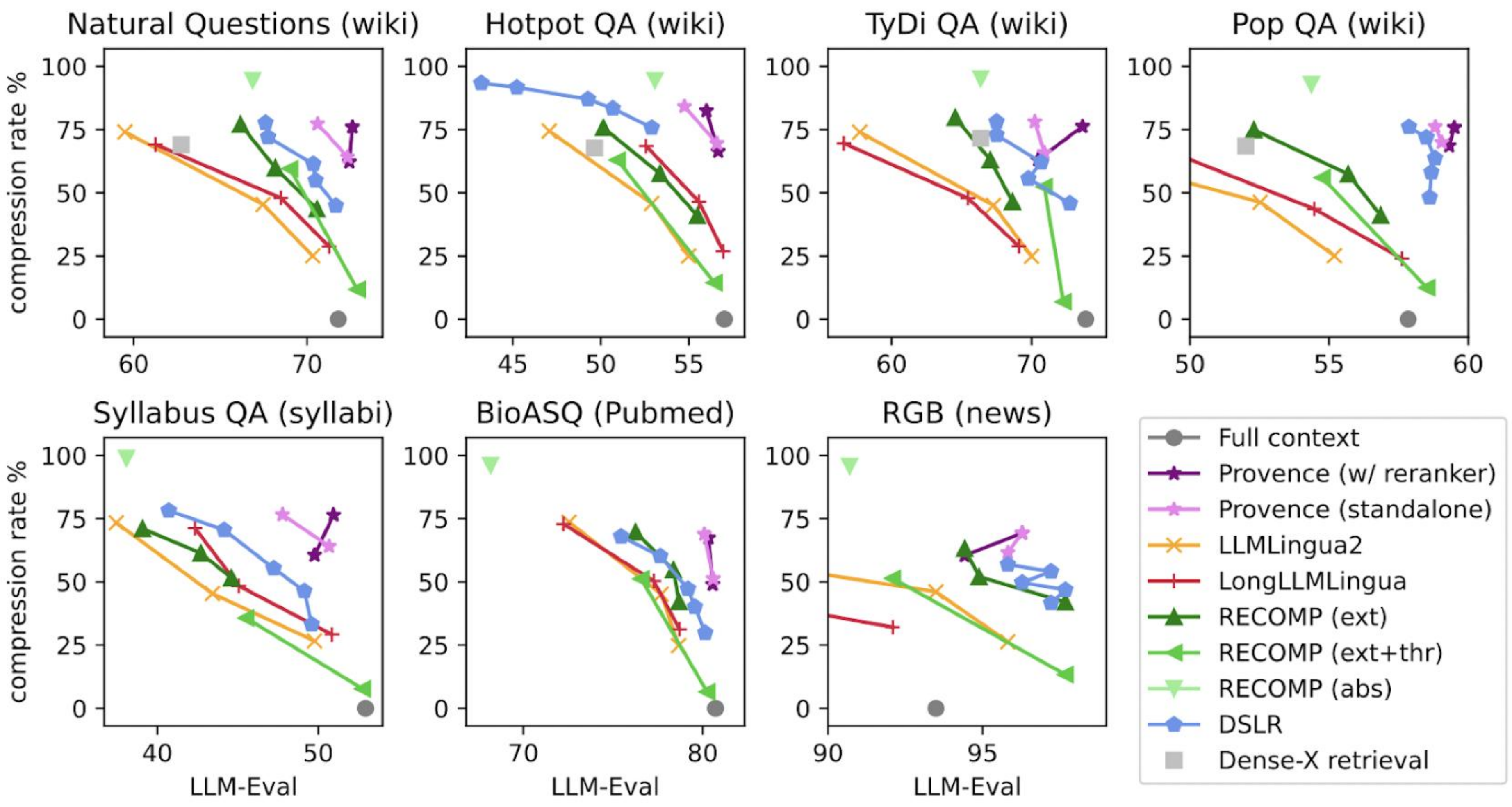
Provence automatically detects the number of sentences to keep, based on a threshold. We found that the default value of a threshold works fine across various domains, but the threshold can be adjusted further to better meet the particular use case needs.

Provence is efficient: Provence provides a compact DeBERTa-based model, and it can be used either as a standalone context pruner or as a unified reranking+context pruning model. In the later case, we incorporate context pruning into reranking, an already existing stage of modern RAG pipelines. The unification of reranking+pruning makes context pruning almost zero cost in the RAG pipeline!

Provence is robust across domains and works out-of-the-box with any LLM and retriever.

Main results

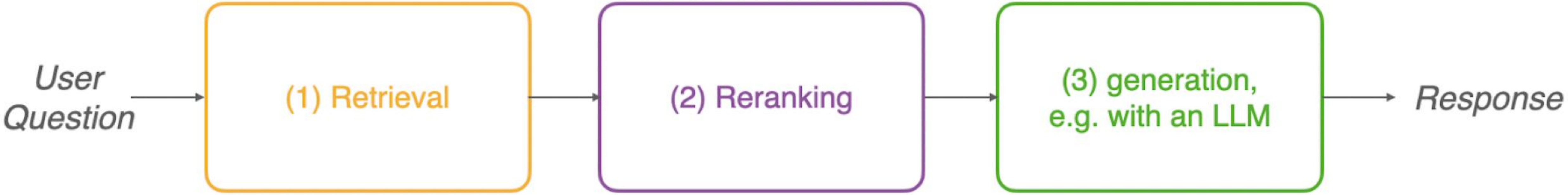
top right corner corresponds to the best model



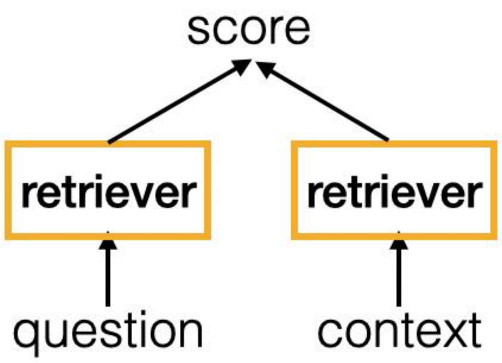
Provence consistently outperforms other approaches, in all domains, and stays on the Pareto front.

Provence is the only model that performs context pruning with little-to-no drop in performance

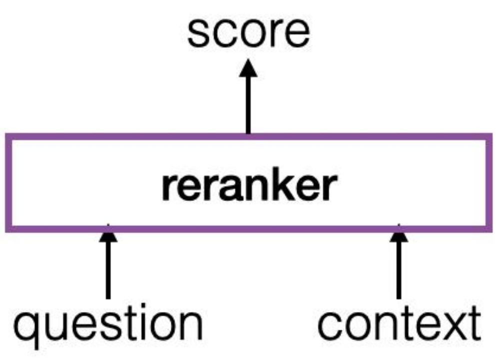
RAG Pipeline



- fast, but less precise first stage of retrieval
- slower, but more precise second stage of retrieval



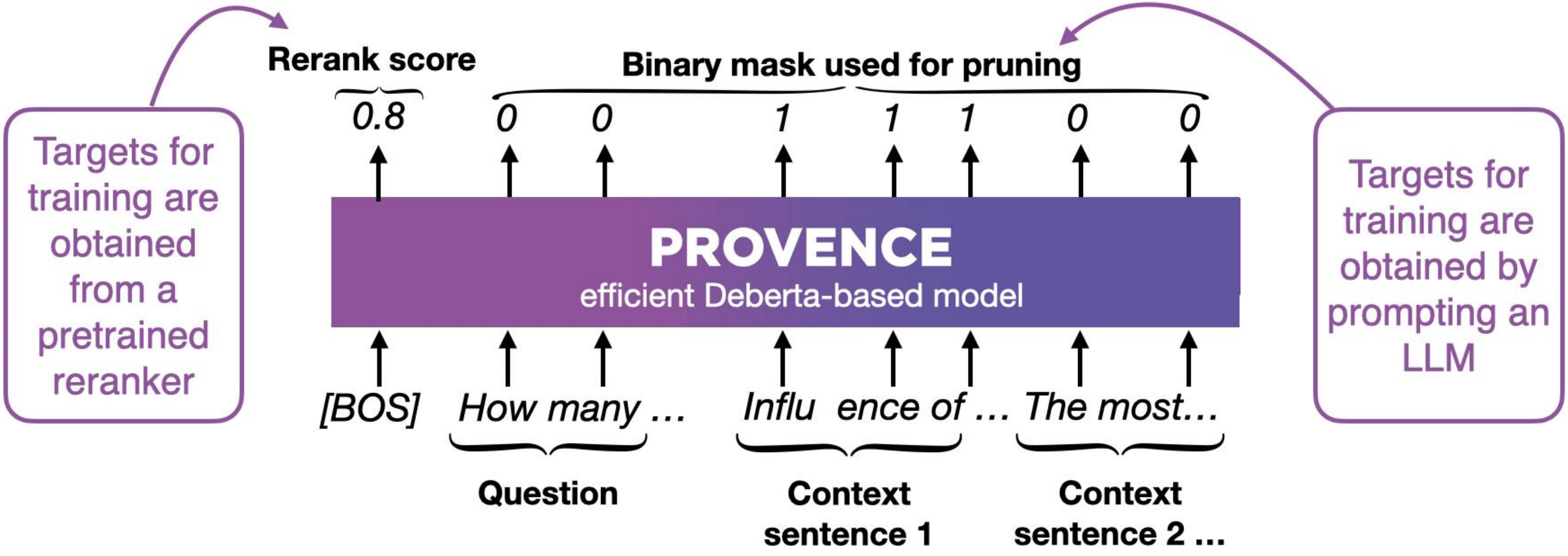
at inference, only need to encode a query → fast (contexts are pre-encoded)



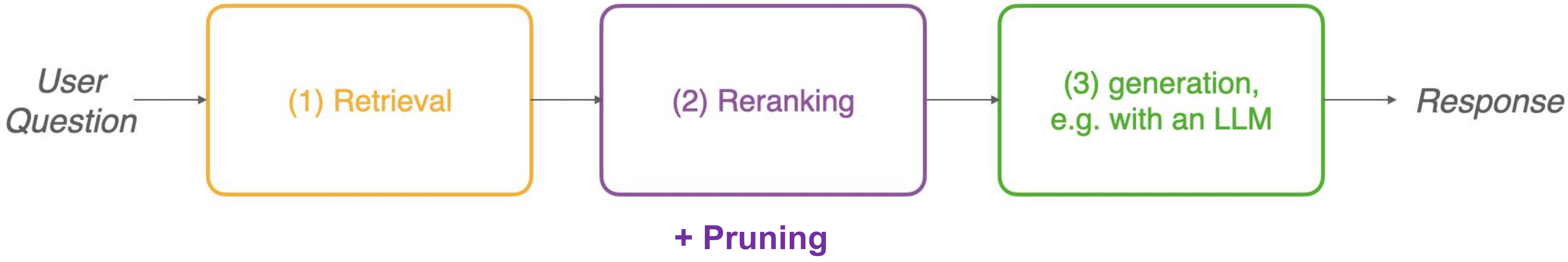
at inference, need to encode each context with a query → slower

Same architecture as Provence!

Architecture: Joint Reranker and Pruner



Online Compression: Hard Compression



A single forward rerank documents and select relevant sentences

Online Compression: Hard Compression

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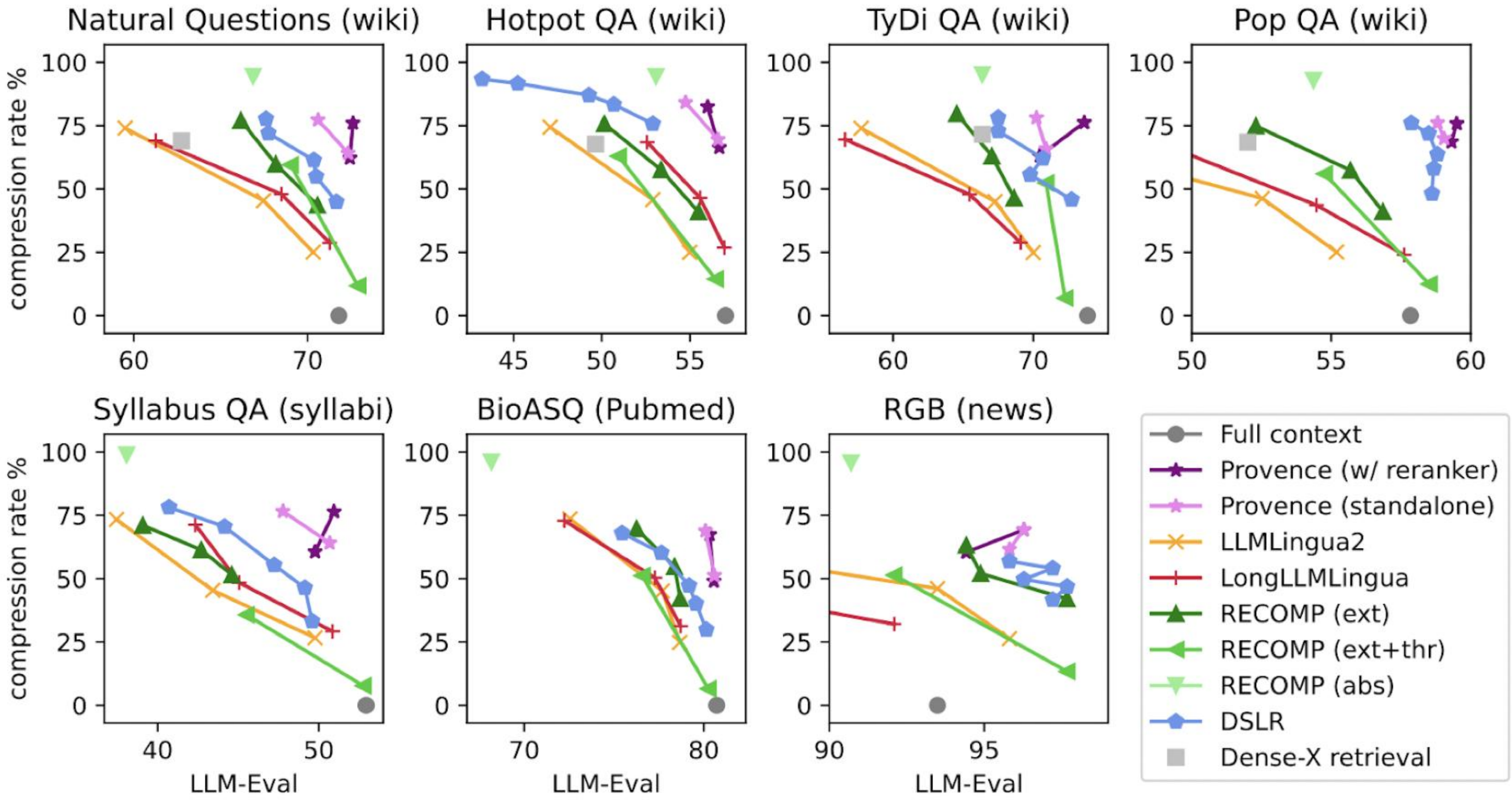

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Online Compression: Hard Compression

Table 2: Time/MFLOPS required for context pruning, 50 samples, with batch size set 1 (1 sample consists of a query and top retrieved documents). Top-5 retrieved documents

Pruner	Time (*s*)	FLOPS
LongLLMLingua, rate=0.5	194	5.400e+16
LLMLingua2, rate=0.5	38	9.822e+15
RECOMP extr., top=2	12	1.037e+15
RECOMP abstr.	499	5.357e+14
DSLR, th=0.5	26	4.391e+15
`Provence (standalone)`	25	7.901e+14

Table 3: Speed up in generation due to compression (`Provence`, 49% compression). Batch sizes 1 or 256.

Generator	*bs* 1	*bs* 256
LLama-2-7B	×1.2	×2
LLama-2-13B	×1.4	×2
SOLAR-10.7B	×1.4	×1.9

A single Forward rerank document and select relevant sentences

Outline

OSCAR